

Propenu Notification Testing Guide

Backend + Firebase push + admin campaign notification process

- User Service
- Property Service
- Firebase FCM
- Postman Testing
- Admin Campaigns

1. What Was Implemented

The notification work now has two paths: automatic event notifications and admin campaign notifications.

Automatic Event Notifications

These are sent by backend business events. The implementation sends push notifications to the property owner or project builder, and also records/sends admin alerts.

Event	Who receives push	Where it is triggered
Buyer/user contacts a property owner	Owner/builder/agent who owns the property, plus admins	<code>property-service/src/services/leadService.ts</code>
Public featured project lead is submitted	Project builder, plus admins	<code>property-service/src/services/publicLeadService.ts</code>
User downloads a featured project brochure	Project builder, plus admins	<code>property-service/src/controller/leadController.ts</code>

Admin Campaign Notifications

Admins can send festival, offer, announcement, or general push notifications with text and optional image.

Audience	Backend target	Use case
<code>all</code>	Every active user with a valid <code>fcmtoken</code>	General festival wishes or platform-wide announcement
<code>builder</code>	Users with builder role	Builder plan offers, project promotions, builder updates
<code>agent</code>	Users with agent role	Agent offers, lead updates, sales campaigns
<code>owner</code>	Users with normal user role	Owner property boost, property posting reminders
<code>user</code>	Users with normal user role	Buyer/renter messages, property recommendations, festival offers

2. Backend Routes Added Or Used

Route	Method	Purpose
<code>/api/users/notifications/save-fcm-token</code>	POST	Saves a user's Firebase device token.
<code>/api/users/notifications/admin/notify/custom</code>	POST	Admin sends campaign push notification with filters and optional image.
<code>/api/users/notifications/admin/feed</code>	GET	Admin reads automatic alerts plus sent campaigns.
<code>/api/users/notifications/admin/feed/seen</code>	POST	Marks admin notification feed as seen.
<code>/api/properties/leads</code>	POST	Creates contact-owner lead and sends automatic push.
<code>/api/properties/leads/project/lead</code>	POST	Creates public featured-project lead and sends automatic push.
<code>/api/properties/leads/project/:projectId/brochure-download</code>	POST	Tracks brochure download and sends automatic push.

3. Required Setup Before Testing

1. Start backend services: gateway, user-service, property-service. Firebase push testing needs user-service and property-service connected to the same Mongo database.
2. Make sure Firebase service account JSON exists and `FIREBASE_KEY_PATH` points to it.
3. Make sure the target account has a valid Firebase `fcmToken`.
4. Use an admin or super admin JWT for admin campaign routes.

Important: FCM push cannot be tested with a fake token. You need a real token generated by the frontend/mobile Firebase SDK.

4. Base URLs For Postman

If testing through gateway, use:

```
http://localhost:4000
```

If testing user-service directly, use:

```
http://localhost:4004
```

If testing property-service directly, use:

```
http://localhost:4003
```

The examples below use the gateway base URL.

5. Postman Authentication

For protected routes, add this header:

```
Authorization: Bearer YOUR_ADMIN_OR_USER_TOKEN
```

For admin campaign routes, the token must belong to a user whose role is `admin` or `super_admin`.

6. Save FCM Token

Before sending any push, save the target user's FCM token.

```
POST http://localhost:4000/api/users/notifications/save-fcm-token
Content-Type: application/json
```

```
{
  "userId": "TARGET_USER_ID",
  "token": "REAL_FIREBASE_FCM_TOKEN"
}
```

Expected response:

```
{
  "message": "Token saved"
}
```

Recommended first test: save the token for your own account, then send an `audience=all` campaign without image.

7. Test Admin Campaign Notification: All Users

Postman setup:

- Method: `POST`
- URL: `http://localhost:4000/api/users/notifications/admin/notify/custom`
- Authorization header: admin token
- Body type: `form-data`

Form-data fields:

```
title: Happy Diwali
body: Wishing you a bright and prosperous festival from Propenu.
audience: all
```

Optional image field:

```
image: select a file in Postman
```

Expected response:

```
{
  "success": true,
  "message": "Notification sent",
  "totalUsers": 10,
  "successCount": 9,
  "failureCount": 1,
  "successUserIds": ["..."],
  "failedUserIds": ["..."],
  "image": "https://...",
  "campaignId": "..."
}
```

8. Test Builder Campaign

```
POST http://localhost:4000/api/users/notifications/admin/notify/custom
```

Body type: `form-data`

```
title: Builder Festival Offer
body: Promote your project with a special festival boost.
audience: builder
```

This targets active users with the builder role and valid `fcmToken`.

9. Test Agent Campaign

```
POST http://localhost:4000/api/users/notifications/admin/notify/custom
```

Body type: `form-data`

```
title: Agent Lead Update
body: New campaign tools are available for agents.
audience: agent
```

10. Test Owner Campaign

```
POST http://localhost:4000/api/users/notifications/admin/notify/custom
```

Body type: `form-data`

```
title: Owner Property Boost
body: Boost your property visibility this festival season.
audience: owner
```

Note: In the current backend, owner maps to the normal `user` role. If later you create a separate owner role, update the audience-role mapping.

11. Test Campaign With Location Filter

Example: send only to agents in Pune, Maharashtra.

```
title: Pune Agent Campaign
body: Special update for agents in Pune.
audience: agent
state: Maharashtra
city: Pune
```

Optional locality:

```
locality: Baner
```

12. Test Campaign To Specific User IDs

Body type: `form-data`

```
title: Selected User Test
body: Testing push notification for selected users.
audience: all
userIds: ["USER_ID_1","USER_ID_2"]
```

In Postman, `userIds` is a text field containing a JSON array string.

13. Test Admin Notification Feed

This endpoint returns automatic admin alerts and manually sent campaigns.

```
GET http://localhost:4000/api/users/notifications/admin/feed
Authorization: Bearer YOUR_ADMIN_TOKEN
```

Expected response shape:

```
{
  "success": true,
  "data": [
    {
      "_id": "...",
      "type": "brochure_downloaded",
      "title": "Brochure Downloaded",
      "body": "...",
      "createdAt": "..."
    },
    {
      "_id": "...",
      "title": "Happy Diwali",
      "body": "...",
      "audience": "all",
      "successCount": 10,
      "failureCount": 0
    }
  ],
  "summary": {
    "total": 2,
    "unread": 2
  }
}
```

14. Mark Admin Feed Seen

```
POST http://localhost:4000/api/users/notifications/admin/feed/seen
Authorization: Bearer YOUR_ADMIN_TOKEN
```

Expected response:

```
{  
  "success": true  
}
```

15. Test Automatic Notification: Contact Owner

This tests the automatic notification sent when a user contacts a property owner.

Requirements:

- The buyer/user must be logged in.
- The property must not belong to the same user.
- The property owner/builder/agent must have a saved `fcmToken`.
- For normal properties, buyer/renter subscription checks still apply.

```
POST http://localhost:4000/api/properties/leads
Authorization: Bearer BUYER_USER_TOKEN
Content-Type: application/json
```

```
{
  "name": "Test Buyer",
  "phone": "9876543210",
  "email": "buyer@example.com",
  "projectId": "PROPERTY_ID",
  "propertyType": "residential",
  "remarks": "I am interested"
}
```

Supported `propertyType` values:

```
featuredprojects
residential
commercial
agricultural
landplots
```

Expected notification title:

```
New Contact Request
```

16. Test Automatic Notification: Public Featured Project Lead

```
POST http://localhost:4000/api/properties/leads/project/lead
Content-Type: application/json
```

```
{
  "projectId": "FEATURED_PROJECT_ID",
```



```
"name": "Public Lead",
"phone": "9876543210",
"email": "lead@example.com",
"message": "Interested in this project"
}
```

Expected notification title:

New Project Lead

17. Test Automatic Notification: Brochure Download

```
POST http://localhost:4000/api/properties/leads/project/FEATURED_PROJECT_ID/brochure-download
Authorization: Bearer USER_TOKEN
```

Expected notification title:

Brochure Downloaded

The project builder and admins receive this notification if they have valid FCM tokens.

18. Where Data Is Stored

Collection	Purpose
users	Stores <code>fcmToken</code> and notification seen timestamps.
notificationcampaigns	Stores admin-sent campaigns, filters, counts, image URL, success/failure users.
usercontentnotifications	Stores per-user records for successful admin campaigns.
platformnotifications	Stores admin alerts generated by backend events such as brochure download and leads.
brochuredownloads	Tracks featured project brochure downloads.
propertyleads	Stores direct contact-owner leads.
publicleads	Stores public featured-project leads.

19. Notifications You Can Receive

Receiver	Notification examples
Admin	Brochure downloaded, new contact request, new project lead, sent campaign history.
Builder	New project lead, brochure downloaded, project contact request, admin campaign.
Agent	New property lead/contact request, admin campaign for agents.
Owner	New property contact request, admin campaign for owners/users.
User/Buyer/Renter	Admin campaign, festival message, platform announcement.

20. Common Errors And Fixes

Error	Meaning	Fix
403 Forbidden	The JWT user is not admin/super_admin.	Login with admin or super_admin account.
No users found	No active users matched the audience/location filters with FCM token.	Check role, city/state/locality spelling, and saved token.
No valid tokens	Matching users exist but do not have fcmToken .	Call save-fcm-token first with a real Firebase token.
Firestore service account file NOT FOUND	Firebase Admin cannot find the service account JSON.	Set FIREBASE_KEY_PATH correctly and restart backend.
messaging/registration-token-not-registered	The FCM token is expired or invalid.	Generate a fresh frontend token and save it again.
This is your own property	User tried to contact their own property.	Use a different user account as buyer/renter.

21. Quick Test Checklist

1. Start gateway, user-service, and property-service.
2. Confirm Firebase service account file path.
3. Login as admin and copy JWT.
4. Save a real FCM token for at least one target user.
5. Send admin campaign with audience=all .
6. Check device/browser push received.

7. Check `GET /api/users/notifications/admin/feed`.
8. Test builder/agent/owner filters.
9. Test brochure download automatic notification.
10. Test contact-owner automatic notification.